

Mphasis

Data Engineer Interview (0–5 YOE)

1 Question: Explain your data pipeline architecture.

What they asked me

“Walk me through the end-to-end pipeline you worked on and how data flows through each layer.”

What I said

I described a real Azure pipeline:

Ingestion

- ADF Copy Activity pulling from SQL + API
- Data stored in **ADLS Bronze** in raw format

Transformation

- Databricks (PySpark) for cleaning and schema normalization
- Silver: Deduplicated, standardized
- Gold: Aggregated business tables

Storage

- Delta Lake for ACID, schema evolution, time travel

Consumption

- Synapse Serverless for queries
- Power BI dashboards

Tips

- Mphasis LOVES the **Bronze** → **Silver** → **Gold** pattern
- Keep your architecture practical, not theoretical

2 Question: Describe your incremental load approach.

What they asked me

“How do you load only new or changed data every day?”

What I said

I used **watermark logic**:

Step 1: Metadata table

Stores last successful timestamp.

Step 2: Filter source

```
SELECT * FROM customer WHERE modified_ts > @watermark
```

Step 3: Apply Delta MERGE

```
MERGE INTO customer_dim t
USING updates s
ON t.customer_id = s.customer_id
WHEN MATCHED THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *
```

Step 4: Update watermark

Tips

- Use the keyword **CDC + MERGE**
 - Avoid saying “we do full load daily”
-

Tips

- Mphasis wants **business + technical rules**
- Always mention error folder + logging

6 Question: ADF Error — Copy Activity Failing

What they asked me

“What are common reasons for Copy Activity failure and how do you debug them?”

What I said

Possible reasons

- Schema mismatch
- File corruption
- Wrong column mapping
- Invalid credentials
- Integration Runtime throttling
- Source API limits

Debug approach

1. Check Activity Output JSON
2. Validate schema
3. Check IR capacity
4. Test connection from Linked Service
5. Retry with smaller batch

Tips

- Mphasis expects **systematic debugging**, not guesswork
-

7 Question: Delta Lake — Implement SCD Type 2

What they asked me

“How do you keep history of records in Delta Lake?”

What I said

Use Delta MERGE with audit columns:

```
MERGE INTO dim_customer t
USING updates u
ON t.customer_id = u.customer_id
WHEN MATCHED AND t.hash <> u.hash THEN
  UPDATE SET t.is_current = false, t.end_date = current_date()
WHEN NOT MATCHED THEN
  INSERT *
```

Used:

- effective_date
- end_date
- is_current
- hash comparison

Tips

- MUST mention hashing for change detection
- SCD2 = critical for Mphasis

8 Question: How do you handle schema drift?

What they asked me

“What happens if a new column appears in the source feed?”

What I said

- Enable schema drift in ADF
- Use Delta's `mergeSchema` option
- Add default values for missing columns
- Update Silver/Gold mapping
- Notify downstream consumers

Tips

- Show both technical + communication steps
- Mphasis values governance

9 Question: Explain your ADLS folder structure design

What they asked me

"How do you structure your data lake folders?"

What I said

```
/bronze/source/table/date=2025-02-10/  
/silver/table/date=2025-02/  
/gold/table/
```

Partitioned by *date*, not high-cardinality fields.

Tips

- Don't say you store everything in one folder
- Partitioning must be clear + logical

10 Question: Why Mphasis?

What they asked me

"Why do you want to join Mphasis?"

What I said

- Strong focus on cloud modernization
- Exposure to BFSI + healthcare pipelines
- Opportunity to work on scalable ETL solutions
- Mphasis framework for governance & automation impressed me
- I want to grow in cloud + big data engineering roles

Tips

- Connect your answer to Mphasis cloud projects
- Mention BFSI/Telecom/Healthcare industry exposure

- Enable schema drift in ADF
- Use Delta's `mergeSchema` option
- Add default values for missing columns
- Update Silver/Gold mapping
- Notify downstream consumers

Tips

- Show both technical + communication steps
- Mphasis values governance

9 Question: Explain your ADLS folder structure design

What they asked me

"How do you structure your data lake folders?"

What I said

```
/bronze/source/table/date=2025-02-10/  
/silver/table/date=2025-02/  
/gold/table/
```

Partitioned by *date*, not high-cardinality fields.

Tips

- Don't say you store everything in one folder
- Partitioning must be clear + logical

10 Question: Why Mphasis?

What they asked me

"Why do you want to join Mphasis?"

What I said